Atlas Oliva: A Secure and Traceable YOLOv8-Based Web Platform for Automatic Olive Tree Detection and Counting from Aerial Imagery

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Abstract. Automated olive tree counting from UAV imagery is a key challenge in precision agriculture for orchard inventory, density estimation and crop management. This paper presents *Atlas Oliva*, a secure web-based platform that integrates a YOLO detector for operational olive tree detection and counting. A harmonized dataset of 741 images was constructed by combining the OliveTreeCrownsDb dataset [9] and the Roboflow Universe Olive Trees Detection v1 dataset [10]. Five YOLO-family models were systematically compared under identical experimental conditions using object-level detection metrics and an image-level binary ROC protocol. YOLOv8s achieved the best overall performance, with a mAP@0.5 of 95.4%, a precision of 93%, a recall of 92%, an F1-score of 92% and a ROC-AUC of 0.9940. For counting, YOLOv8s achieved an MAE of 3.60 trees per image on the full-resolution 1×1 test subset and an overall MAE of 5.17 trees per image on the 132-image multi-scale test subset. The platform integrates JWT authentication, role-based access control, audit logging, secure password hashing, file validation and geographic visualization via Leaflet. The results demonstrate the feasibility of combining deep learning with application-level security controls for operational olive grove monitoring.

Keywords: Olive tree detection · YOLOv8 · Precision agriculture · UAV imagery · JWT authentication · FastAPI · Deep learning

1 Introduction

Remote sensing has become a central component of precision agriculture because aerial and satellite observations can support field monitoring, crop mapping and

management decisions at fine spatial resolution [3]. The rapid adoption of UAV platforms has further increased the availability of high-resolution RGB imagery, making it possible to observe individual tree crowns and within-field variability at parcel scale [5].

Morocco is one of the world's leading olive oil producers, with more than one million hectares of olive groves. In this context, automated counting and localization of olive trees from aerial imagery is essential for inventory, density estimation, yield planning and the early identification of missing or abnormal plants [7, 8]. Manual tree counting on aerial images is time-consuming and sensitive to observer variability. Deep learning has shown strong potential in agricultural image analysis by automatically learning visual features from annotated data [4, **?**]. The YOLO family of detectors is particularly attractive for operational systems because it detects multiple objects in a single forward pass and provides a practical compromise between accuracy and inference speed [1].

However, a deployable agricultural decision-support tool requires more than a detection model. Existing UAV analysis pipelines typically focus on offline detection or segmentation, whereas operational use also requires authenticated access, controlled uploads, analysis history, role separation and traceability. Atlas Oliva is positioned in this gap: it connects an olive-tree detector to a traceable and secured web workflow suitable for parcel-level analysis.

The main contributions of this work are:

- A harmonized olive-tree detection dataset combining OliveTreeCrownsDb [9], available at `https://data.mendeley.com/datasets/xym8rd2srf/3`, and the Roboflow Universe Olive Trees Detection v1 dataset [10], available at `https://universe.roboflow.com/saras-workspace-ju8qv/olive-trees-c9fji/1`, totaling 741 annotated images.
- A systematic comparison of five YOLO-family models—YOLOv8n, YOLOv8s, YOLOv8m, YOLO11n and YOLO26n—trained under identical experimental conditions on the combined dataset, using mAP@0.5, precision, recall, F1-score and image-level ROC-AUC.
- An end-to-end tiled inference pipeline for large aerial images, based on 640×640 pixel tiling, confidence filtering and non-maximum suppression.
- A web platform integrating the selected YOLOv8s model with application-level security mechanisms, including JWT authentication, role-based access control, audit logging, file format validation and geographic visualization.

2 Related Work

2.1 Remote Sensing and Deep Learning for Agriculture

Remote sensing in precision agriculture has evolved from coarse field observation toward high-resolution monitoring of soil, vegetation and crop status [3]. Deep learning has accelerated this evolution by enabling automatic feature extraction for classification, detection and segmentation tasks in agricultural imagery [4]. UAV-based remote sensing is particularly relevant because low-altitude imagery

captures canopy structure and local spatial variability that may not be visible in medium-resolution satellite data [5].

Tree-crown detection is a well-studied problem in forestry and agroforestry. Weinstein et al. [6] showed that semi-supervised deep learning can detect individual tree crowns in RGB imagery. In olive orchards, Safonova et al. [7] used Mask R-CNN for olive tree crown and shadow segmentation from UAV imagery. Ye et al. [8] proposed an olive crown extraction approach based on UAV visible images and U^2-Net. These studies show that performance depends strongly on acquisition conditions, spatial resolution, crown size, soil background and the geographic region.

2.2 Olive Datasets and Object Detection

The development of olive-specific datasets is essential for reliable model training and benchmarking. OliveTreeCrownsDb [9] provides high-resolution UAV imagery and fine-grained annotations for olive tree crown detection and segmentation. The Roboflow Universe Olive Trees Detection v1 dataset [10] provides an additional open-source corpus with bounding-box annotations suitable for YOLO training. Compared with prior work that focuses mainly on offline detection or segmentation performance, Atlas Oliva extends olive-tree detection to a complete operational environment that combines automatic detection with user authentication, analysis history, audit logging and geospatial visualization.

3 Materials and Methods

3.1 Dataset Preparation

The experimental dataset combines two public sources with complementary characteristics. The first source is the 1x1 subset of OliveTreeCrownsDb [9], available at `https://data.mendeley.com/datasets/xym8rd2srf/3`, containing 46 high-resolution UAV images with fine-grained bounding-box annotations in YOLO format. The second source is the Roboflow Universe Olive Trees Detection v1 dataset [10], available at `https://universe.roboflow.com/saras-workspace-ju8qv/olive-trees-c9fji/1`, containing 695 open-source annotated images in YOLOv8 format. Class names were harmonized to `olive_tree` and both sources were exported in a unified YOLO format. The combined dataset therefore contains 741 images. Table 1 summarizes this composition.

Table 1: Composition of the final training dataset.

Source	Images	Format
OliveTreeCrownsDb 1x1 subset [9]	46	YOLO
Olive Trees Detection v1 [10]	695	YOLO
Total	**741**	YOLO

The two sources differ in acquisition height, spatial resolution, crown appearance, soil color and illumination conditions, which may introduce domain shift effects. The dataset was split into training, validation and test subsets before any model training. Near-duplicate images derived from the same UAV acquisition sequence were assigned to the same subset to prevent data leakage between training and evaluation. The effective split was 463 images for training (62.5%), 146 for validation (19.7%) and 132 for testing (17.8%), as summarized in Table 2.

Table 2: Train/validation/test split of the combined dataset.

Subset	Images	Proportion
Training	463	62.5%
Validation	146	19.7%
Test	132	17.8%
Total	**741**	100%

3.2 Model Training

Five YOLO-family models were compared under identical experimental conditions: YOLOv8n, YOLOv8s, YOLOv8m, YOLO11n and YOLO26n. All models were trained on the same combined dataset of 741 images with a fixed input size of 640×640 pixels, a batch size of 8, 50 epochs and a patience of 10 epochs for early stopping. The AdamW optimizer was selected automatically by Ultralytics. Training was performed on Google Colab with a Tesla T4 GPU using Ultralytics 8.4.57 [2].

The YOLOv8 architecture comprises a CSPDarknet backbone for multi-scale feature extraction, a feature pyramid neck for feature aggregation and an anchor-free detection head. For counting, each retained bounding box after non-maximum suppression is interpreted as one detected olive tree instance. For images larger than 640×640 pixels, the tiled inference pipeline divides the image into 640×640 pixel tiles, runs inference independently on each tile, and projects the detections back into the coordinate system of the original image. The confidence threshold for inference was set to 0.25 and the NMS IoU threshold to 0.4. Global non-maximum suppression across tile boundaries is identified as future work.

Figure 1 illustrates the complete methodology pipeline.

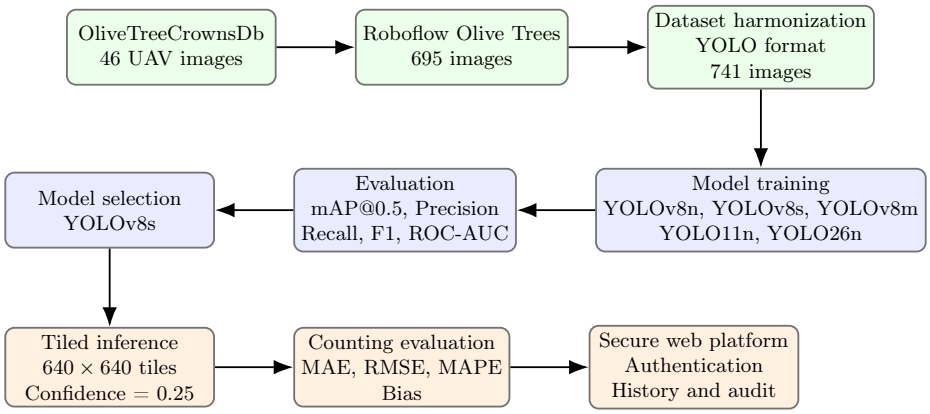


Fig. 1: Complete methodology pipeline of Atlas Oliva, including dataset construction, model training, evaluation, tiled inference, counting assessment and secure web deployment.

3.3 Secure Platform Architecture

Atlas Oliva follows a layered web architecture illustrated in Figure 2. The frontend is implemented with HTML, CSS, JavaScript and Leaflet.js, enabling the user to authenticate, select a Moroccan geographic region, upload an aerial image, trigger detection and visualize bounding boxes with geographic markers. The backend is implemented with FastAPI and exposes REST endpoints for authentication (`/auth/login`, `/auth/register`, `/auth/logout`), detection (`/detect`), analysis history (`/analyses`), audit logs (`/audit-logs`) and system health checks (`/health`). Analysis results and audit events are persisted in SQLite.

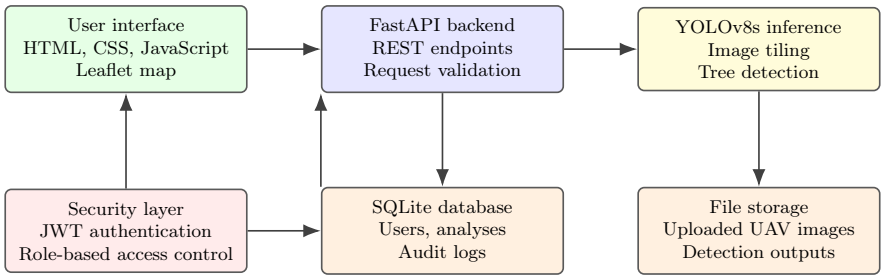


Fig. 2: Readable architecture of Atlas Oliva. The user interface sends image analysis requests to the FastAPI backend, which invokes the YOLOv8s inference module. Results are stored as files, while metadata, users, analyses and audit logs are persisted in SQLite. The security layer protects access through JWT authentication and role-based access control.

Security mechanisms are integrated at the application level. User passwords are hashed with PBKDF2-HMAC-SHA256 with a random 128-bit salt and 120 000 iterations. After successful login, the server issues a JWT signed with HS256 containing the user identifier, role, issue time, expiry time and a unique token identifier (JTI) [12]. Logout revokes the token by storing its JTI in a dedicated SQLite table; subsequent requests with the same token are refused with HTTP 401. Role-based access control separates analyst and administrator privileges: analysts can access only their own counting history, whereas administrators can access the complete analysis history and audit log. Uploaded files are restricted to JPG, JPEG and PNG extensions. All database queries use parameterized statements to prevent SQL injection. Security validation followed the OWASP ASVS guidelines [13].

4 Experimental Evaluation

4.1 Detection Metrics

The evaluation uses mAP@0.5, precision, recall, F1-score, inference speed and image-level ROC-AUC. All five models were trained and evaluated under identical conditions on the same 741-image dataset to ensure a fair comparison.

For the image-level binary ROC protocol, the score assigned to each image is the maximum detection confidence produced by the model. ROC-AUC measures the ability to discriminate between olive-tree presence and background at image level; it complements but does not replace object-level mAP and Precision-Recall evaluation.

Table 3 reports the full comparison of the five evaluated models. Inference speed corresponds to a 640×640 input on a Tesla T4 GPU.

Model	mAP@0.5	Precision	Recall	F1	Speed (ms)	ROC-AUC
YOLOv8n	94.2%	91%	89%	90%	6.7	0.9719
YOLOv8s	**95.4%**	**93%**	**92%**	**92%**	**4.5**	**0.9940**
YOLOv8m	94.5%	92%	91%	91%	10.0	0.7679
YOLO11n	94.1%	91%	89%	90%	5.0	0.9286
YOLO26n	91.7%	88%	86%	87%	12.3	0.6228

Table 3: Quantitative comparison of the five YOLO-family models on the combined 741-image dataset. YOLOv8s achieves the best performance across the evaluated metrics and is selected as the production model for Atlas Oliva.

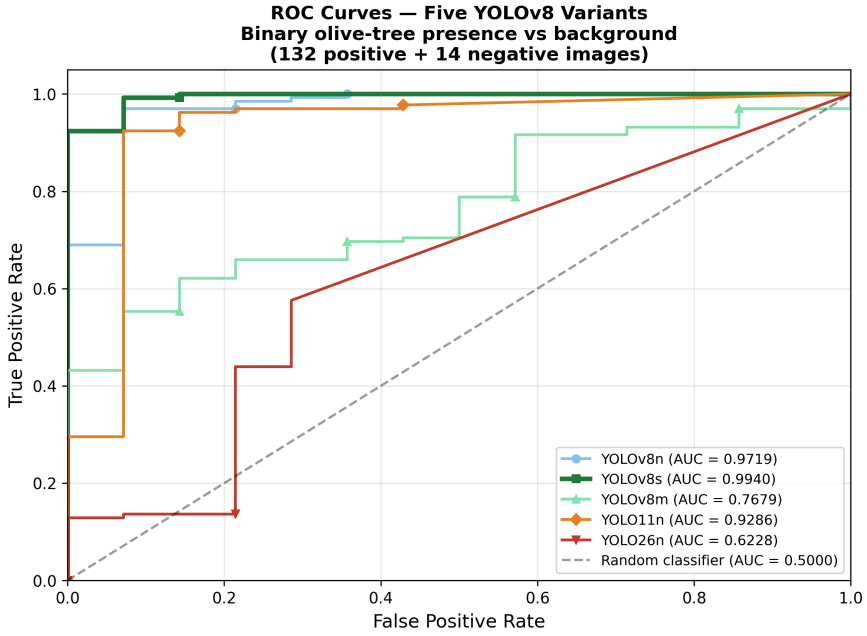


Fig. 3: ROC curves of the five evaluated models under the binary olive-tree presence versus background protocol. YOLOv8s achieves the highest ROC-AUC of 0.9940.

4.2 Comparison with State of the Art

Table 4 compares Atlas Oliva with existing methods for olive tree detection and counting from aerial imagery. Methods are compared on the basis of detection approach, dataset size, reported accuracy metric and whether application-level security is integrated.

Table 4: Comparison with existing olive tree detection and segmentation methods from aerial imagery.

Method	Architecture	Dataset	Best reported metric	Task	Security
Safonova et al. [7]	Mask R-CNN	OTCSS UAV	F1 = 1.000 on shadow segmentation at 3 cm/pixel	Segmentation	No
Ye et al. [8]	U^2-Net	Custom UAV	RMSE = 4.78 and MRE = 14.27% for crown area extraction	Crown extraction	No
Weinstein et al. [6]	RetinaNet	NEON RGB	Precision = 0.61 and recall = 0.69 on mixed-species tree crowns	Tree detection	No
Atlas Oliva (ours)	**YOLOv8s**	**741 images**	**mAP@0.5 = 95.4%, MAE = 3.60 trees/image on** 1×1 **images**	**Detection**	**Yes**

Atlas Oliva achieves a mAP@0.5 of 95.4% with YOLOv8s and a ROC-AUC of 0.9940. Direct comparison with Safonova et al. [7] and Ye et al. [8] is limited by differences in evaluation protocol, since these works focus on segmentation or crown extraction rather than bounding-box detection. A key distinction of Atlas Oliva is the integration of application-level security mechanisms, including JWT authentication, role-based access control and audit logging, which are absent from the compared methods.

4.3 Ablation Study

Table 5 reports the impact of the tiling strategy on counting accuracy. All experiments use the same YOLOv8s model and confidence threshold of 0.25 on the 132-image test subset.

Table 5: Ablation study: effect of tiled inference on olive tree counting accuracy. MAE and RMSE are reported in trees per image.

Configuration	MAE (trees/image)	RMSE (trees/image)
YOLOv8s without tiling	35.46	65.08
YOLOv8s with tiling (640×640)	**5.17**	**8.62**
Relative improvement	*85.4%*	*86.8%*

The tiling strategy reduces the MAE from 35.46 to 5.17 trees per image and the RMSE from 65.08 to 8.62 trees per image. This substantial improvement confirms that direct full-image inference at 640×640 pixels causes significant undercounting on high-resolution UAV images where individual olive crowns are small relative to the image dimensions. Tiling preserves the apparent size of

individual crowns within each tile, enabling the model to apply the detection scale it was trained on and substantially improving counting accuracy.

4.4 Model Selection

The systematic comparison identifies **YOLOv8s** as the best-performing model across the evaluated metrics, achieving a mAP@0.5 of 95.4%, a precision of 93%, a recall of 92%, an F1-score of 92% and a ROC-AUC of 0.9940. Among all evaluated models, YOLOv8s offers the best compromise between detection accuracy and inference speed at 4.5 ms per image. YOLOv8m achieved comparable mAP (94.5%) but at a higher inference cost (10.0 ms). YOLO26n obtained the lowest mAP (91.7%) on this dataset, which may be explained by its stronger optimization for latency rather than dense object detection accuracy on high-resolution UAV imagery. YOLOv8s is therefore selected as the production model integrated into the Atlas Oliva platform.

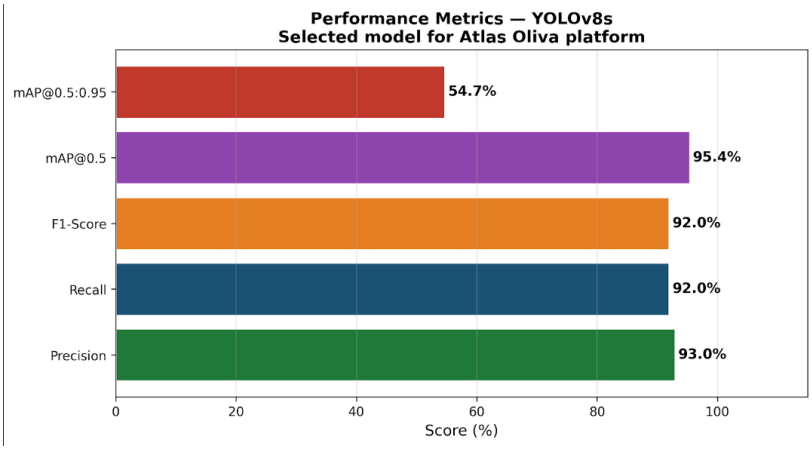


Fig. 4: Performance metrics of YOLOv8s on the 132-image test subset: precision, recall, F1-score, mAP@0.5 and mAP@0.5:0.95.

4.5 Counting Evaluation

The counting evaluation quantifies the difference between the number of detected olive trees and the number of annotated trees on the 132-image test subset containing 4145 ground-truth annotations. The following metrics are reported:

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n} |N_{\mathrm{detected},i} - N_{\mathrm{real},i}| \tag{1}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} (N_{\mathrm{detected},i} - N_{\mathrm{real},i})^2} \tag{2}$$

$$\mathrm{MAPE} = \frac{100}{n} \sum_{i=1}^{n} \frac{|N_{\mathrm{detected},i} - N_{\mathrm{real},i}|}{N_{\mathrm{real},i}} \tag{3}$$

$$\mathrm{Bias} = \frac{1}{n} \sum_{i=1}^{n} (N_{\mathrm{detected},i} - N_{\mathrm{real},i}) \tag{4}$$

The 95% confidence interval for the MAE is:

$$\mathrm{CI}_{95\%} = \mathrm{MAE} \pm 1.96 \, \frac{s_{|e|}}{\sqrt{n}} \tag{5}$$

where $s_{|e|}$ denotes the standard deviation of the per-image absolute errors.

Table 6: Counting results of YOLOv8s on the 132-image multi-scale test subset. MAE, RMSE and Bias are reported in trees per image.

Subset	Images	GT	Pred.	MAE	RMSE	MAPE	Bias
1×1	20	3034	3100	3.60	4.32	2.59%	+2.18
3×3	40	760	1085	8.93	13.97	82.02%	+6.54
6×6	40	236	212	3.60	5.44	69.42%	−1.72
9×9	32	115	5	3.44	3.69	94.43%	−3.44
Overall	132	4145	4402	5.17	8.62	69.17%	+1.95

The positive overall bias of +1.95 trees per image indicates a moderate tendency toward over-detection. The 95% confidence interval for the overall MAE is [3.99, 6.35]. The high MAPE values observed for the 3×3, 6×6 and 9×9 subsets are primarily explained by the very small ground-truth counts in cropped tile images: an absolute error of only a few detections produces a large relative percentage error when the reference count is small. These values should therefore be interpreted together with MAE and RMSE, and not as standalone indicators of model weakness. The 1×1 subset, which contains full-resolution images with larger tree counts, achieves a stable MAPE of 2.59% and an MAE of 3.60 trees per image, confirming that YOLOv8s performs well on complete high-resolution UAV images.

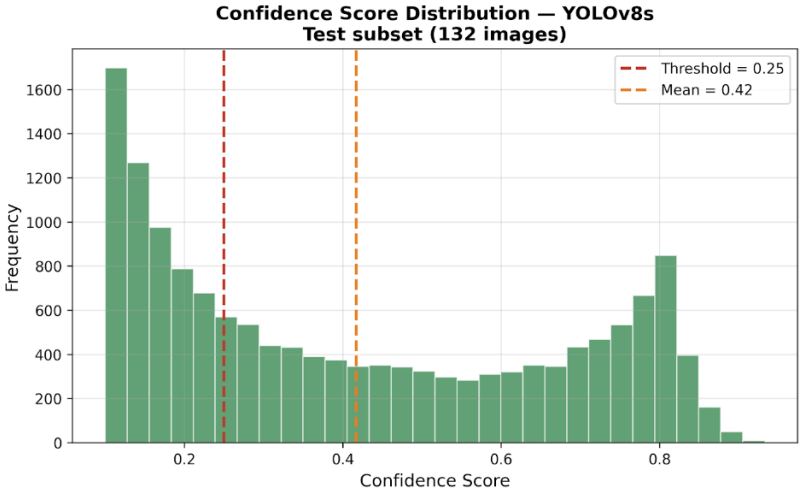


Fig. 5: Distribution of detection confidence scores of YOLOv8s on the test subset. The red dashed line indicates the inference threshold of 0.25.

4.6 Qualitative Detection Examples

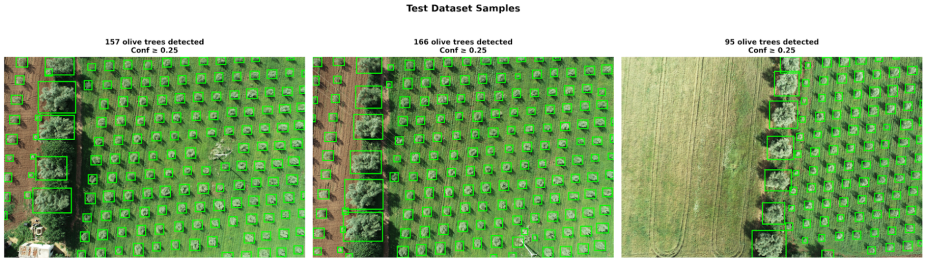


Fig. 6: Representative olive tree detection results with YOLOv8s on test images from the combined dataset. Green bounding boxes indicate detected olive tree crowns.

4.7 External Image Evaluation

To qualitatively assess generalization, YOLOv8s was tested on an external satellite image of an olive orchard in the Marrakech region not included in any training or evaluation subset. The system detected 131 olive trees with a mean confidence of 61% using 640×640 tiling. These qualitative results should not be used as quantitative counting accuracy measures since no complete ground-truth annotation is available for the external images. They are presented exclusively for

qualitative generalization assessment. The lower mean confidence compared with the test set confirms a domain shift caused by differences in spatial resolution, soil appearance, crown morphology and sensor characteristics between the training data and the external satellite image.



(a) External image before detection.



(b) External image after YOLOv8s detection.

Fig. 7: Qualitative external evaluation on a satellite image from the Marrakech region. The first image shows the original image before detection, while the second image shows the YOLOv8s detection result with 131 detected olive trees and a mean confidence of 61%. No complete ground truth is available; this result is presented for qualitative generalization assessment only.

4.8 Security Validation

Functional security tests were performed on the Atlas Oliva prototype. The tests confirmed that protected endpoints reject requests without JWT authentication, invalid credentials return HTTP 401, revoked tokens are refused after logout, and analyst and administrator privileges are correctly separated. In particular, analysts can access only their own analyses through `/analyses`, while administrators can access the complete analysis history and audit logs. Invalid file formats are refused, and SQL injection attempts are neutralized through parameterized queries. These tests validate the main application-level security controls of the prototype, but they should not be interpreted as a full production security audit. The permissive CORS configuration is a non-conformity that must be corrected before production deployment.

5 Discussion

The experimental results confirm that YOLOv8s is well-suited for olive crown detection when the input imagery is consistent with the training domain. Among the five evaluated models, YOLOv8s achieved the highest mAP@0.5 of 95.4% and ROC-AUC of 0.9940, while maintaining a competitive inference speed of 4.5 ms per image. YOLOv8m achieved comparable mAP (94.5%) but at a higher inference cost (10.0 ms), making it less suitable for operational deployment. YOLO26n obtained the lowest mAP (91.7%) on this dataset. This result may be explained by its stronger optimization for deployment efficiency, making it less adapted to dense olive crown detection in high-resolution UAV imagery.

The ROC-AUC results reveal an interesting divergence between object-level mAP and image-level discrimination ability. YOLOv8m achieves a comparable mAP@0.5 of 94.5% but a notably lower ROC-AUC of 0.7679, suggesting a tendency to produce false detections on background images despite good localization performance on positive images. Similarly, YOLO26n achieves the lowest ROC-AUC of 0.6228, only marginally above random discrimination. YOLOv8n and YOLO11n achieve intermediate ROC-AUC values of 0.9719 and 0.9286, respectively, while YOLOv8s dominates with a ROC-AUC of 0.9940.

The counting evaluation shows that YOLOv8s detects 4402 trees against 4145 annotated trees on the 132-image test subset, yielding an overall MAE of 5.17 trees per image, an RMSE of 8.62 trees per image and a mean bias of +1.95 trees per image, with a 95% confidence interval for the MAE of [3.99, 6.35]. The positive bias indicates a moderate tendency toward over-detection. The high MAPE values observed for small-scale tiles are primarily explained by the very low ground-truth counts in cropped subimages, where small absolute errors produce large relative percentages. The 1×1 subset, which contains full-resolution images and larger tree counts, achieves a stable MAPE of 2.59% and an MAE of 3.60 trees per image.

The external Marrakech satellite test provides qualitative evidence of generalization across geographic contexts, but the lower mean confidence highlights a

domain shift caused by differences in sensor characteristics, spatial resolution, soil color, illumination and crown morphology. Complete ground-truth annotation of these external images is required before quantitative use.

Several limitations remain. First, the tiled inference pipeline does not yet apply global non-maximum suppression across tile boundaries, which may produce duplicate detections in high-density border zones. Second, the imbalance between the two dataset sources may introduce domain shift effects that influence generalization performance. Third, the permissive CORS configuration must be corrected before production deployment. Finally, full reproducibility requires the exact dataset split, Ultralytics version, trained weights, confidence and NMS thresholds, random seed and execution configuration.

6 Conclusion and Future Work

This paper presented Atlas Oliva, a secure web platform for automated olive tree detection and counting from aerial imagery. A harmonized dataset of 741 images was constructed by combining OliveTreeCrownsDb [9] and the Roboflow Universe Olive Trees Detection v1 dataset [10]. Five YOLO-family models were systematically compared under identical experimental conditions. **YOLOv8s achieved the best overall performance with a mAP@0.5 of 95.4%, a ROC-AUC of 0.9940 and an inference speed of 4.5 ms per image**, and was selected as the production model for the Atlas Oliva platform.

The counting evaluation showed an MAE of 3.60 trees per image on the full-resolution 1×1 test subset and an overall MAE of 5.17 trees per image on the 132-image multi-scale test subset. Beyond detection accuracy, Atlas Oliva integrates application-level security mechanisms, including JWT authentication, token revocation, role-based access control, audit logging, password hashing and file validation. In particular, analysts can access only their own counting history, while administrators can access the complete analysis history and audit log.

Future work will focus on: (1) implementing global non-maximum suppression across tile boundaries to eliminate duplicate detections; (2) expert annotation of external satellite images for quantitative generalization evaluation; (3) source-aware domain-shift analysis between the two dataset sources; (4) production hardening through HTTPS, restrictive CORS, rate limiting and secret rotation; (5) replacement of the current SQLite persistence layer with a production-grade database system for multi-user deployment at scale; and (6) investigation of overlap-based tiling with configurable stride to further reduce duplicate detections at tile boundaries.

Acknowledgements The authors thank the Ecole Nationale des Sciences Appliquees, Universite Cadi Ayyad, for the academic context of this project.

Disclosure of Interests The authors have no competing interests to declare that are relevant to the content of this article.

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